

Statistics.

- Statistics: It is the study of collection, organization and interpretation of data to draw useful conclusion.

→ Statics: Physics → Forces in equilibrium.

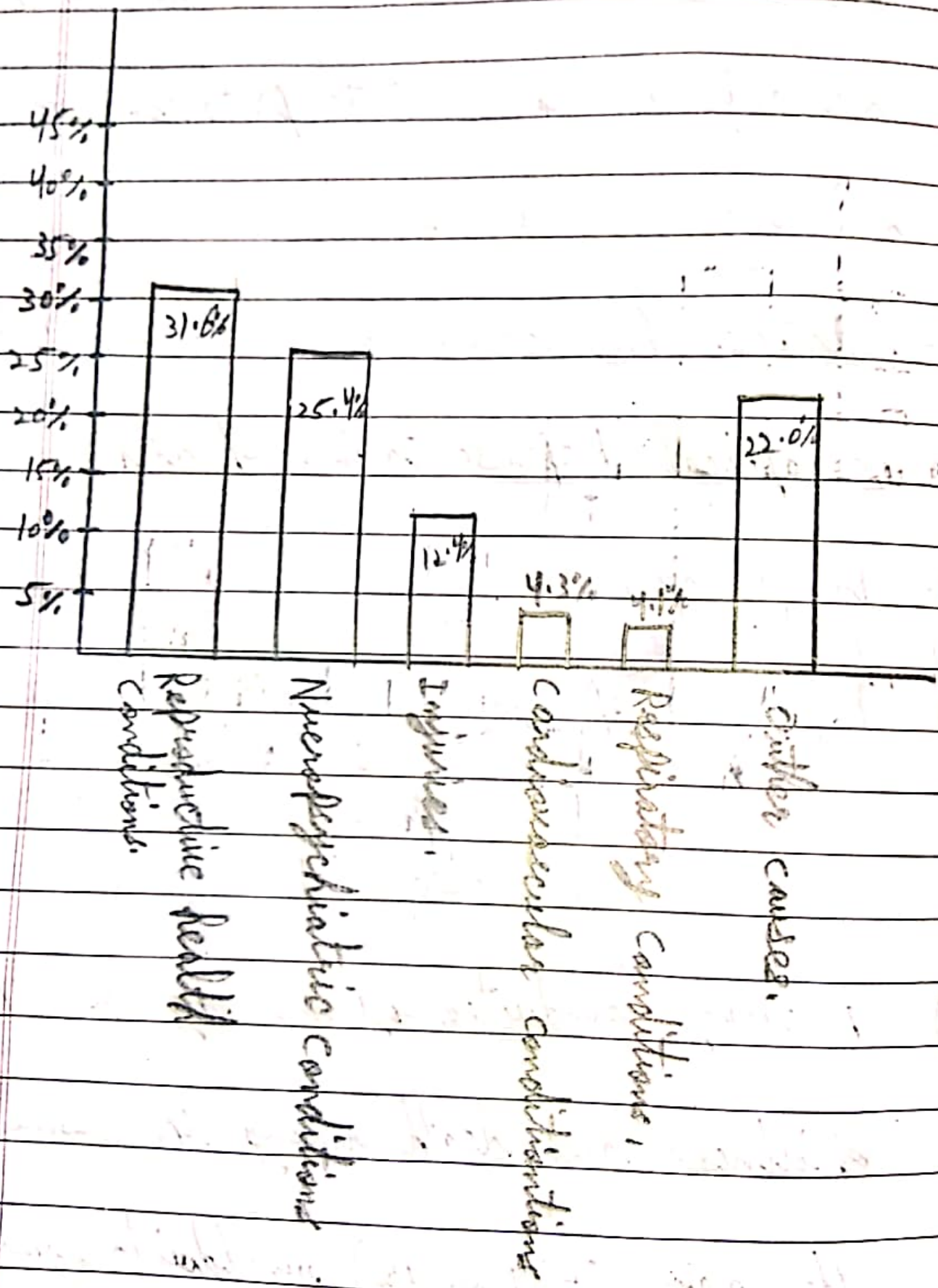
⇒ Graphical Representation of data:-

- 1) Bar Graph.
- 2) Histograms
- 3) Frequency polygon → Number of times an observation occurs.

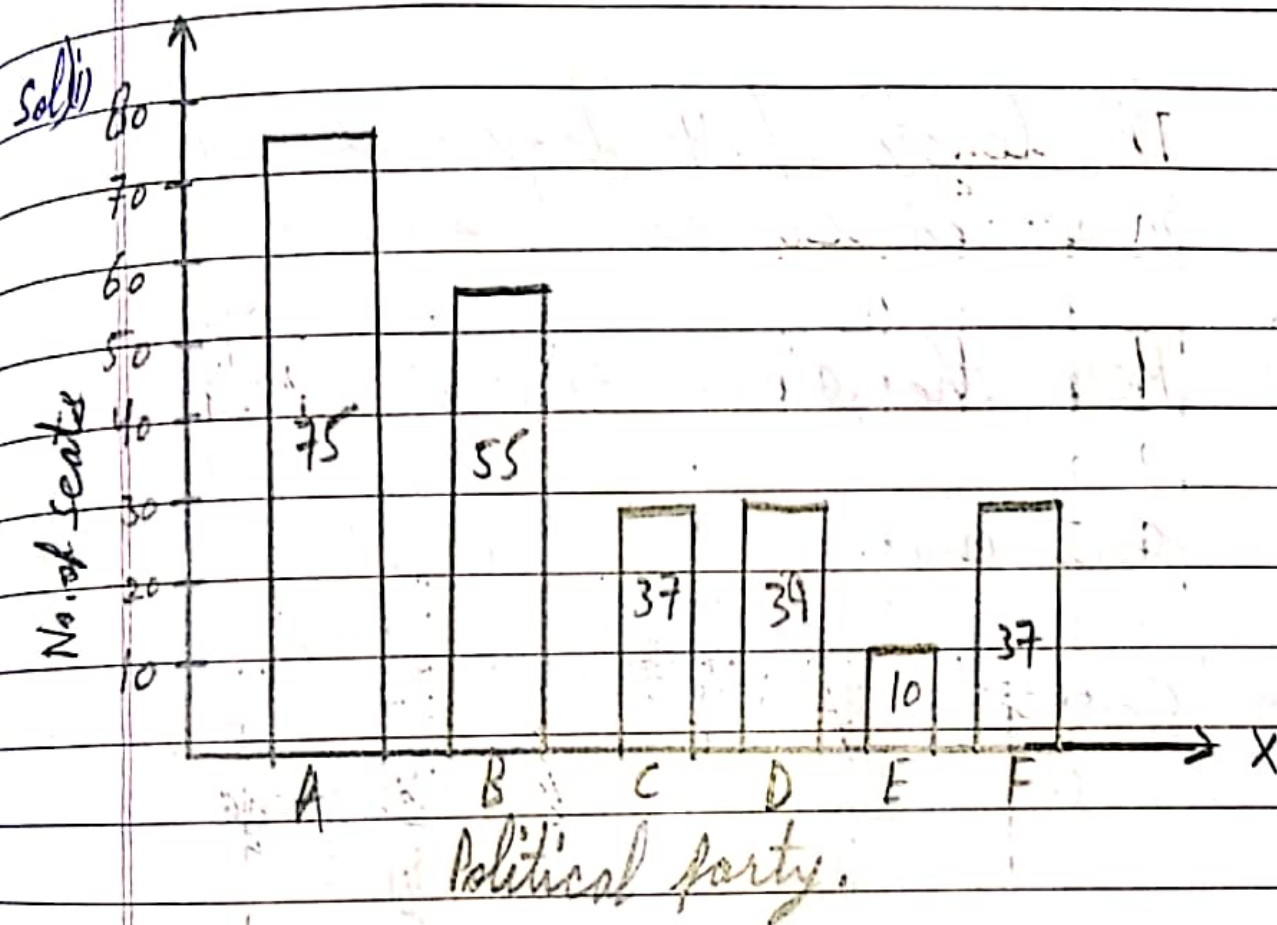
Exercise = 11.1

Q1) A survey conducted by an org. for the cause of illness and death among the women between the ages 15-44 (in yrs) worldwide, found the following figures (in %).

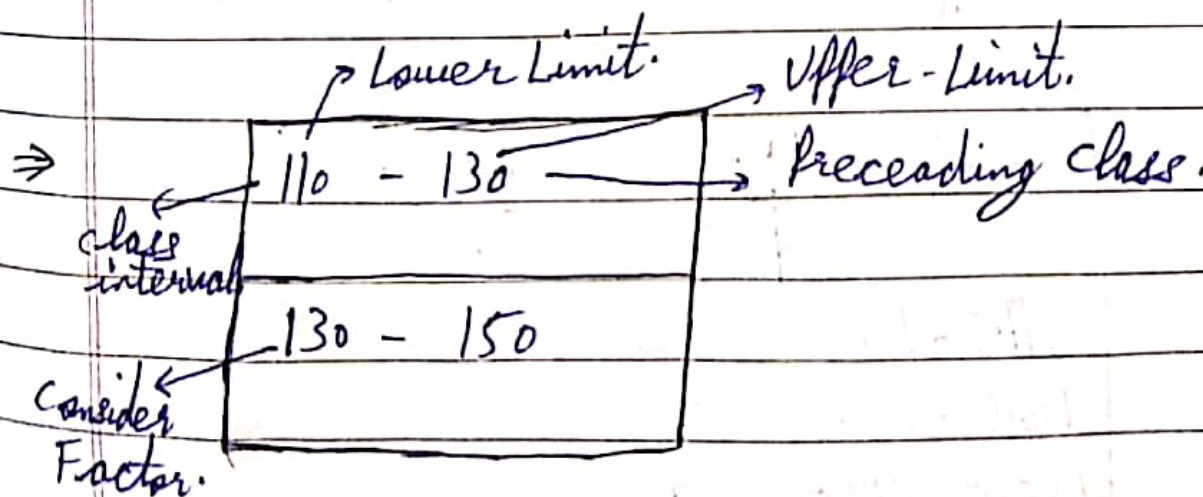
Sol:ii scale: 1cm = 5%



Q3: Given below are the seats won by different political parties in the polling outcome of a state assembly elections. ?



(ii) Party A has won maximum number of seats.



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★ → $\text{Correction} = \frac{\text{Value of Lower Limit factor of a class} - \text{Value of upper limit of Preceding class}}{2}$

Q4: The length of 40 leaves of a plant are measured correct to ~~one~~ one millimetre.

Sol) Here the given distribution is not continuous.

⇒ $\text{Correction factor} = \frac{\text{Lower Limit} - \text{Upper limit of preceding class}}{2}$

$$= \frac{127 - 126}{2} = \frac{1}{2} = 0.5$$

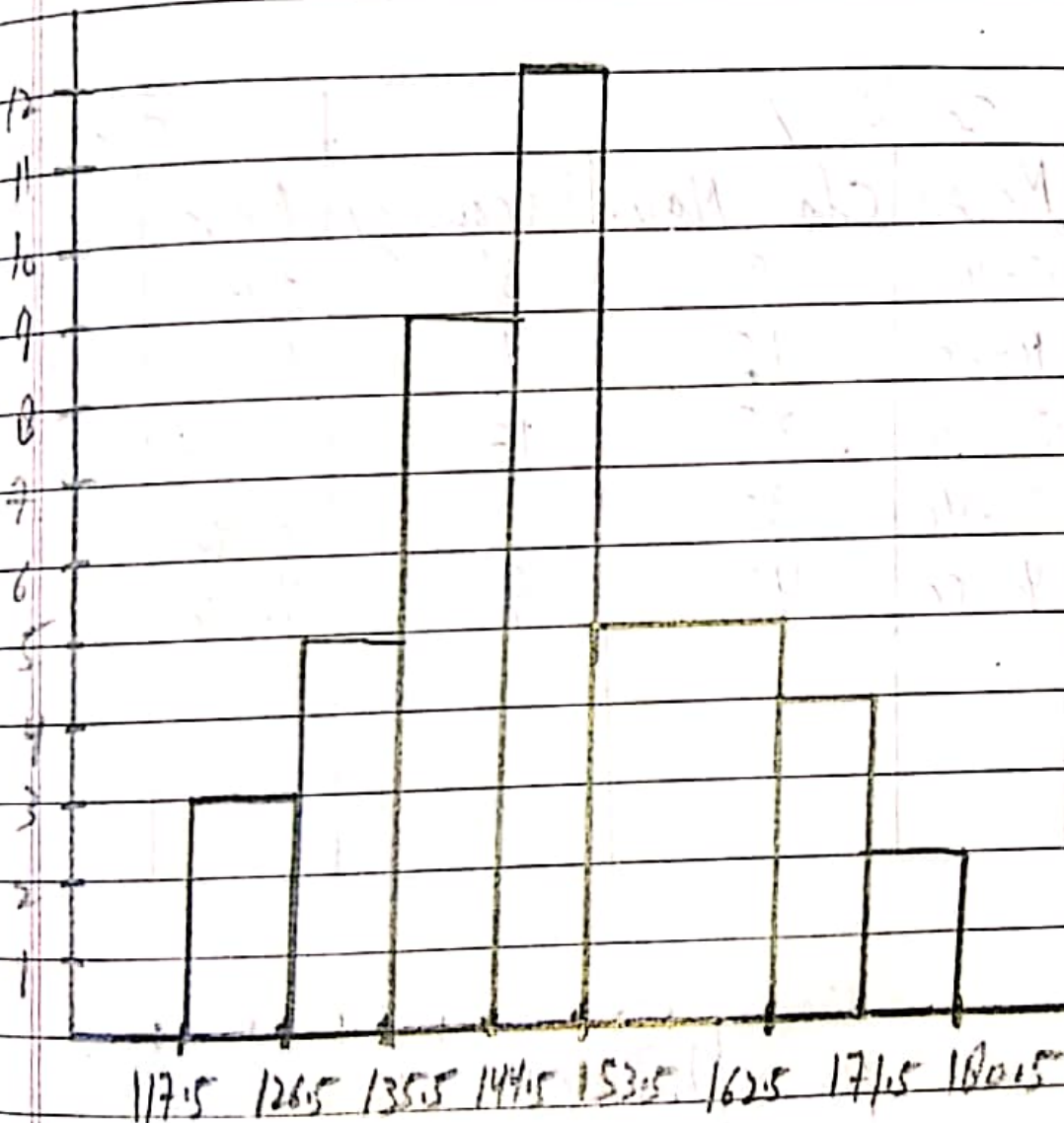
⇒

Length (mm)	No. of Leaves
117.5 - 126.5	3
126.5 - 135.5	5
135.5 - 144.5	9

144.5-153.5	12
153.5-162.5	5
162.5-171.5	4
171.5-180.5	2

1 leave = 0.5 cm

Frequency → No. of leaves



Length (mm) →

Ans) ii) Frequency Polygon.

Ans) iii) No.

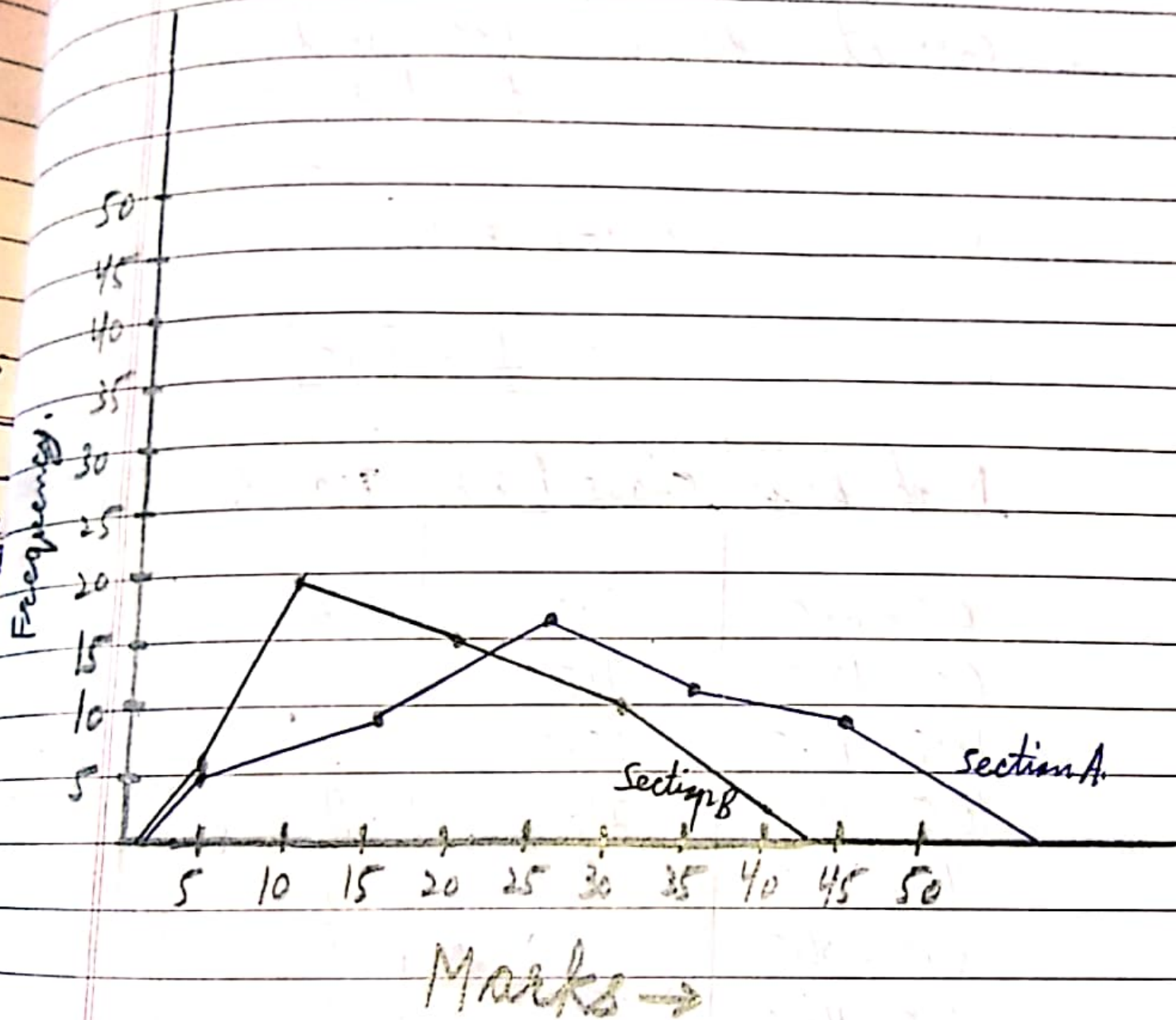
$\Rightarrow$ class Mark
 $=$ Mid point of class
 $= \frac{\text{Upper Limit} + \text{Lower Limit}}{2}$

Q6: Represent the marks of the students of both the sections graph by two frequency polygons?

Section A			Section B		
Marks	Class Mark	Frequency	Marks	Class Mark	Frequency
0-10	5	3	0-10	5	5
10-20	15	9	10-20	15	19
20-30	25	17	20-30	25	15
30-40	35	12	30-40	35	10
40-50	45	9	40-50	45	1

P.T.O

• scale 1cm = 5 marks.



Q7: The runs scored by two teams A and B on first 60 balls in a cricket match are - ?

Ans- The given distribution is not continuous.

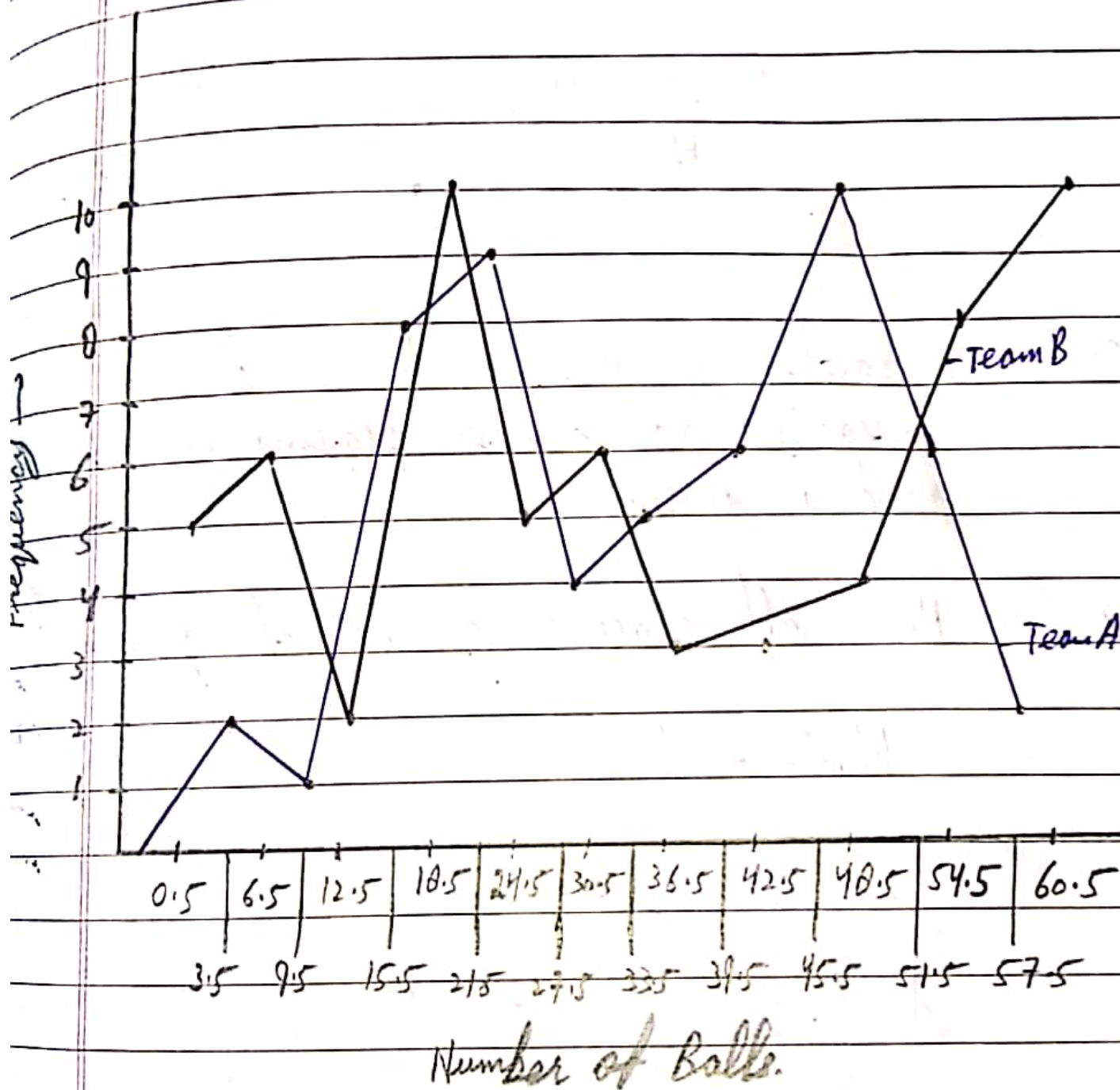
⇒ correction factor = $\frac{\text{Lower limit} - \text{Upper Limit of preceding class}}{2}$

$$= \frac{7 - 6}{2}$$

$$= \frac{1}{2} = \boxed{0.5}$$

No. of balls	class Mark	Team A	Team B
0.6 - 6.5	3.5	2	5
6.5 - 12.5	9.5	1	6
12.5 - 18.5	15.5	8	2
18.5 - 24.5	21.5	9	10
24.5 - 30.5	27.5	4	5
30.5 - 36.5	33.5	5	6
36.5 - 42.5	39.5	6	3
42.5 - 48.5	45.5	10	4
48.5 - 54.5	51.5	6	8
54.5 - 60.5	57.5	2	10

⇒ Scale 1cm = 1 run.



Minimum class width, ($W_{\min} = 2$)

⇒ Scale 1 cm = 4 people.

